

Benchmark 2 Study Guide

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Major Topics to Review:

- **Measures of Central Tendency**
 - Understand the definitions and differences between mean, median, mode, and their sensitivity to extreme scores.
 - Practice identifying which measure is most or least affected by outliers.
- **Frequency Distributions**
 - Review what histograms, bar charts, and pie charts display.
 - Know how to read and interpret a histogram and what it tells about a data set.
- **Distribution Shapes**
 - Review skewness (right and left skewed distributions).
 - Be able to relate mean, median, and mode positions in skewed distributions.
- **Graphical Data Representation**
 - Learn which graph is appropriate for displaying categorical or numerical data (bar chart, pie chart, histogram, boxplot, scatterplot).
 - Practice reading and describing different graphs.
- **Measures of Variability**
 - Understand the concepts of range, variance, and standard deviation.
 - Know which statistics measure spread or dispersion in a distribution.
- **Boxplots and Outliers**
 - Learn how boxplots display distribution and how they reveal the presence of outliers.
- **Levels of Measurement**
 - Review the four main levels: nominal, ordinal, interval, and ratio.
 - Know which levels are required for different calculations (e.g., mean, standard deviation).

- **Jamovi Applications**

- Practice using Jamovi to create and interpret tables and graphs based on levels of measurement.
- Be able to select suitable visualizations for categories vs continuous data.

- **Matching Terms and Definitions**

- Review definitions for terms such as variance, median, outlier, frequency, and how to match them accurately.

- **Reasoning and Judgment**

- Avoid thinking only about formulas, focus on understanding concepts, interpreting visuals, and recognizing correct application.

Study Strategies:

- Revisit your textbook chapters and Jamovi modules that cover: distribution shapes, central tendency, variability, and graph creation.
- Practice interpreting example graphs and outputs from Jamovi, focus on explaining what you see, not just calculation steps.
- Use concept mapping to link the terms to their definitions.
- Practice classifying variable types, and choosing the correct statistic or graph for each.
- Aim for speed and clarity, review with practice quizzes or flashcards if possible.